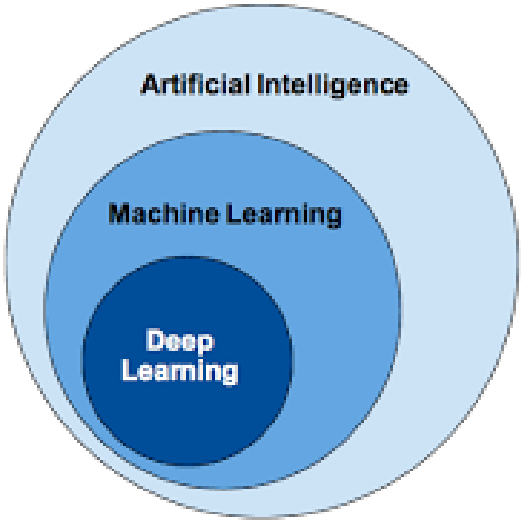
Topic Name: Machine Learning



The concept of learning

X1	X2	Profit
2	1	4000
3	3	9000
1	2	5300
3	2	7100
0	3	6000

X1: # of branches in Amman

X2: # of branches in Dubai

Profit = (X1 + 2 X2) * 1000

relation … equation … model

Prediction

- The importance of learning is to predict the result for new data.

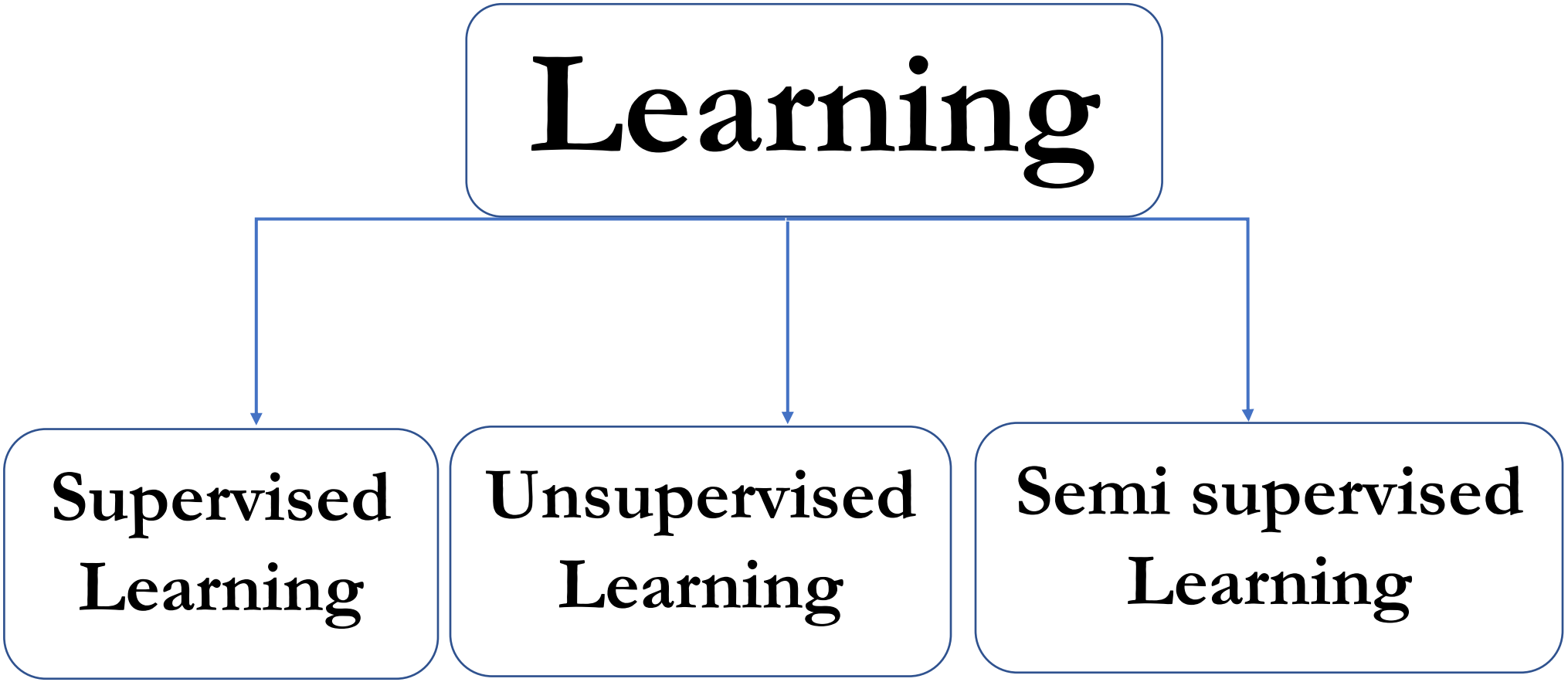
- Example: If someone plans to open 2 branches in Amman, and 3 branches in Dubai. What is the estimated profit according to learned model?

The concept of Features, labels, and samples

Example

age	city	test1	test2	male/female	Cold/Flu
41	Amman	109	30	m	0
22	Salt	124	27	f	0
11	Zarqa	89	32	f	1
57	Madaba	140	37	m	0
44	Amman	99	28	m	1
65	Irbid	112	22	f	0

Types of Learning

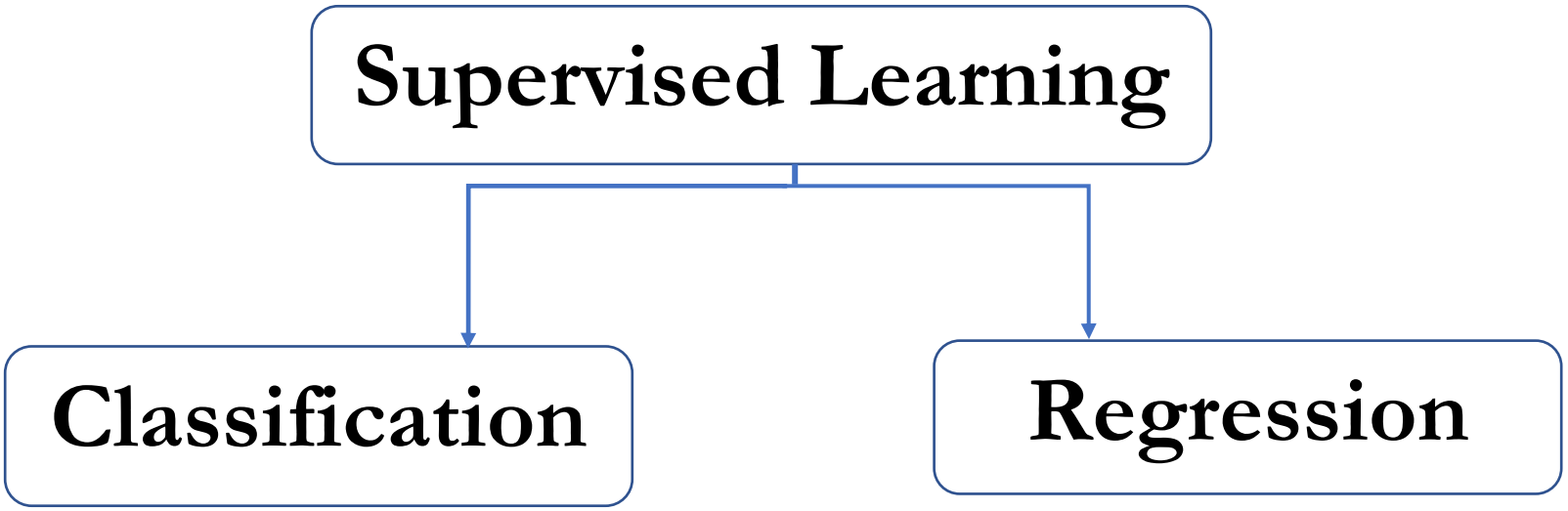


features + (labels/values/outputs)  Supervised learning

features with no (labels/values/outputs)  Unsupervised learning

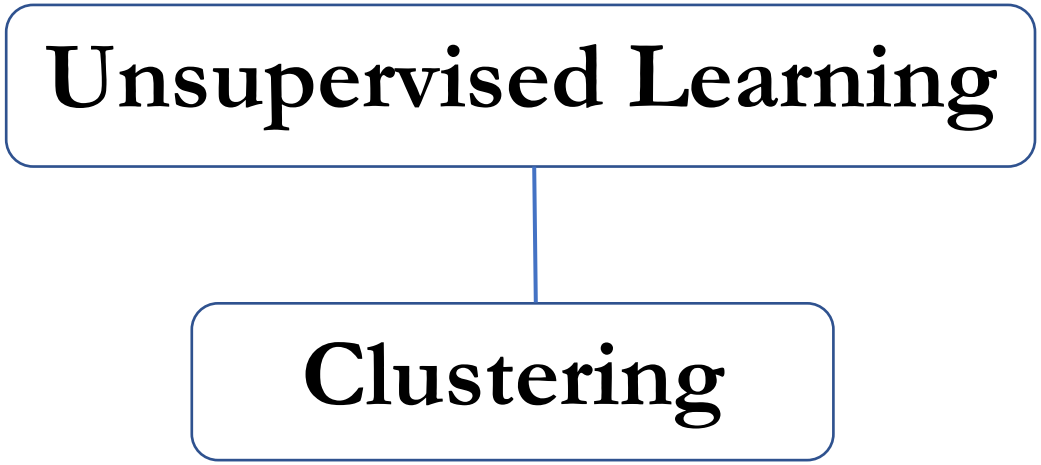
Some samples have labels, and some have not  Semi supervised learning

Classification, Regression, and Clustering



Classification → discrete values

Regression → continuous range


Unsupervised Learning
Clustering

Examples

- **Patients**

Age	city	test1	test2	male/female	Cold/Flu
41	Amman	109	30	m	0
22	Salt	124	27	f	0
11	Zarqa	89	32	f	1
57	Madaba	140	37	m	0
44	Amman	99	28	m	1
65	Irbid	112	22	f	0

Category: classification

- **Pictures of animals**

Width	height	color	background	Object
--	--	--	--	dog
--	--	--	--	cat
--	--	--	--	cat
--	--	--	--	mouse
--	--	--	--	dog
--	--	--	--	mouse

Category: classification

- **Houses**

Area	# of rooms	city	Street No.	Price
--	--	--	--	42,500
--	--	--	--	39,000
--	--	--	--	55,000
--	--	--	--	19,400
--	--	--	--	67,230
--	--	--	--	23,800

Category: regression

Again… The concept of learning

Regression case:

X1	X2	Profit
2	1	4000
3	3	9000
1	2	5300
3	2	7100
0	3	6000

Algorithm-A

Profit = (X1 + 2 X2) * 1000

Algorithm-B

Profit = (X1 + 2 X2) * 1000 + 200

Classification case

x1	x2	x3	Y
2	4	1	1
7	5	5	0
4	6	2	0
3	9	6	1
11	2	7	1

- Algorithm-A gives the following model:

$$Y = \begin{cases} 0 \text{ , if } x1 + x2 + x3 - 2 < 15 \\ 1 \text{ , otherwise} \end{cases}$$

- Algorithm-B gives the following model:

$$Y = \begin{cases} 0 \text{ , if } x1 + x2 + x3 < 20 \\ 1 \text{ , otherwise} \end{cases}$$

Training and Testing

- Suppose that you have a dataset of 1,000 samples (rows).

- Split it into two groups; one for training and the other for testing.

- The features and labels (inputs and outputs) of the training data are given to the algorithm. (why?)

- For testing, the algorithm is given just the inputs. (why?)

Have a look on the following dataset.

	x1	x2	Y
1	2	4	6
2	7	5	12
3	4	6	11
4	3	9	13
5	11	2	13
6	5	2	9
7	4	3	7
8	8	2	10
9	3	6	10
10	1	2	3

The following rows are randomly selected for training (1,2,4,5,7,9,10).

Training

	x1	x2	Y
1	2	4	6
2	7	5	12
3	4	6	11
4	3	9	13
5	11	2	13
6	5	2	9
7	4	3	7
8	8	2	10
9	3	6	10
10	1	2	3

* Algorithm-A: Y = x1 + x2

* Algorithm-B: Y = x1 + x2 + 1

Testing

	x1	x2	Y
1	2	4	6
2	7	5	12
3	4	6	11
4	3	9	13
5	11	2	13
6	5	2	9
7	4	3	7
8	8	2	10
9	3	6	10
10	1	2	3

Algorithm-A: Y = x1 + x2
row#3: Y = 4+ 6 = 10
row#6: Y = 5 + 2 = 7
row#8: Y = 8 + 2 = 10

Actual value	Algorithm-A value (predicted)
11	10
9	7
10	10

Algorithm-B: Y = x1 + x2 + 1
row#3: Y = 4+ 6 + 1 = 11
row#6: Y = 5 + 2 + 1= 8
row#8: Y = 8 + 2 +1 = 11

Actual value	Algorithm-B value (predicted)
11	11
9	8
10	11

Accuracy of classification

$$Accuracy = \frac{Number\ of\ correct\ predictions}{\text{Total number of predictions}}$$

Example:

Actual value	Algorithm value (predicted)
1	1
1	1
0	1
1	1
0	0
0	1
1	1
0	0
0	0
1	1

Accuracy = 8 / 10 = 80%

Accuracy of regression

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} (Predicted_i - Actual_i)^2}{N}}$$

Example:

Actual value	Algorithm value (Predicted)
4	6
6	4
8	9
12	12

$\sum = ((6\text{-}4)^2 + (4\text{-}6)^2 + (9\text{-}8)^2 + (12\text{-}12)^2) = (4 + 4 + 1 + 0) = \mathbf{9}$

RMSE $= \sqrt{9/4} = \sqrt{2.25} = \mathbf{1.5}$

Traditional algorithms ~ Machine learning algorithms

Traditional algorithms (e.g. Sorting, search, …)

Inputs are known and the algorithm is specific.

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Supervised M.L algorithms

Inputs and outputs are known (given)

The algorithm will learn (derive the model… find the relation)